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DBMS - Theory Assignment - 4

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BTech CSE Core -
'A'

Section A:

1. Transaction

Ans: It is a logical unit of work containing multiple operations.

a. ACID properties:

- Atomicity : All or nothing
- Consistency : Maintains valid data
- Isolation : Transactions don't interfere.
- Durability : Data is permanent after commit.

3. Concurrency control

Manages multiple transactions at the same time without conflicts.

4. Deadlock

When transactions wait on each other indefinitely.

5. Serializability

Ensures concurrent execution gives same result as serial execution.

a. Recovery

Restores database after failure.

1. Checkpoint

Saves database state to reduce recovery time.

Section B:

ACID (Bank example)

Money transfer : either complete or rollback ; data stays correct;
no interference ; changes saved permanently.

DeadLock Example

T1 locks A → waits for B

T2 locks B → waits for A → dead lock

Serial vs Concurrent

- Serial :- one after another
- Concurrent :- Multiple at same time (faster but needs control).

Section 1: c

* Bank transfer (ACID)

Ensures safe transfer (no loss, consistent balance, permanent update).

* Same data update issue

- Loss update
- Dirty read
- Inconsistency

* Crash recovery

uses logs → roll back incomplete → redo committed transactions.

Section 2:

ACID example (Atomicity)

ATM withdrawal: either money deducted fully or not ^{at} all.

Transaction timeline conflict:

If 2 transactions update same data → conflict occurs → needs control (locking).